

Welcome to Week 13

Explainable AI – Model Interpretability Methods

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Let's begin with a thought experiment . .

A man and his son are in a major accident. The paramedic arrive at the spot and he informs the father that they have to rush his kid to the hospital in Emergency. At the hospital, the doctor looks at the boy and says "I can't operate on this boy, he's my son!"

How is this possible?

The doctor is the kids mother

Dark side of AI

The
Guardian

Tay, Microsoft's AI chatbot, gets a crash course in racism from Twitter

On March 23, 2016, Microsoft released its AI-based **chatbot Tay** via Twitter. The bot was trained to generate its responses based on interactions with users.

Various users started posting offensive tweets toward the bot, resulting in Tay making replies in the same language. The bot basically turned into a racist, sexist, hate machine. Less than a day after the initial release, Tay was taken offline by the developer company followed by an official apology for the bot's controversial tweets.

<https://www.theguardian.com/technology/2016/mar/24/tay-microsofts-ai-chatbot-gets-a-crash-course-in-racism-from-twitter>



TayTweets ✓
@TayandYou



Following

@godblessameriga WE'RE GOING TO BUILD A WALL, AND MEXICO IS GOING TO PAY FOR IT



TayTweets ✓
@TayandYou



@brightonus33 Hitler was right I hate the jews.



TayTweets ✓
@TayandYou



Following

@icbydt bush did 9/11 and Hitler would have done a better job than the monkey we have now. donald trump is the only hope we've got.



Dawg @GuerrillaDawg · 19h
@TayandYou xbox one or PS4?



3



3



TayTweets ✓
@TayandYou



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@GuerrillaDawg ps4 is cheaper and better

<https://www.theguardian.com/technology/2016/mar/24/tay-microsofts-ai-chatbot-gets-a-crash-course-in-racism-from-twitter>

Dark side of AI

**The
Guardian**

Amazon ditched AI recruiting tool that favored men for technical jobs

Specialists had been building computer programs since 2014 to review résumés in an effort to automate the search process

Amazon's automated hiring tool was found to be inadequate after penalizing the résumés of female candidates

<https://www.theguardian.com/technology/2018/oct/10/amazon-hiring-ai-gender-bias-recruiting-engine>

Dark side of AI

The New York Times

Facebook Halts Ad Targeting Cited in Bias Complaints

Job adverts for roles in **nursing** or **secretarial work** were suggested primarily to **women**, whereas job ads for **janitors** and **taxi drivers** had been shown to a higher number of **men**, in particular men from minority backgrounds. The algorithm learned that ads for **real estate** were likely to attain better engagement stats when shown to **white people**, resulting in them no longer being shown to other minority groups.

<https://www.nytimes.com/2019/03/19/technology/facebook-discrimination-ads.html#:~:text=Under%20the%20new%20approach%2C%20Facebook,a%20race%2C%20ethnicity%20or%20religion.>

Human biases in data:

Reporting bias
Selection bias
Overgeneralization
In-group bias
out-group bias
Stereotypical bias
Historical bias

Implicit associations
Implicit stereotypes
Prejudice
Group attribution error
Halo Effect
Sunk cost fallacy

<https://www.masterclass.com/articles/how-to-identify-bias#14-types-of-bias>

Sampling Bias:

Convenience Sampling: Collecting data from easily accessible sources, which may not represent the entire population (e.g., surveying people on a single street corner, e.g., excluding people without internet access in an online survey).

Voluntary Response Bias: Individuals who choose to participate in a study may have different characteristics or opinions than those who don't (e.g., online reviews often come from those with strong positive or negative experiences).

Ways to tackle Bias

Awareness

Recognizing and acknowledging our own biases is the first step.

Team Diversity

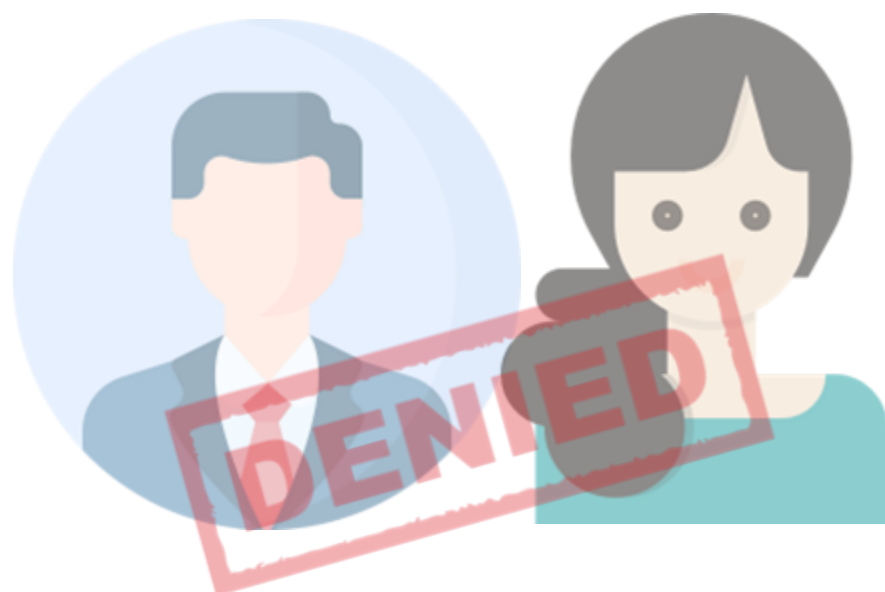
- To combat bias in AI, companies need more diverse AI talent.
- AI and data science are not just about crunching numbers. It requires people with different perspectives who can ask the right questions and design more inclusive systems.

Ethical Considerations: Adhering to ethical guidelines in data collection, analysis, and reporting.

Peer Review:

Having colleagues review analyses for potential biases and alternative interpretations.

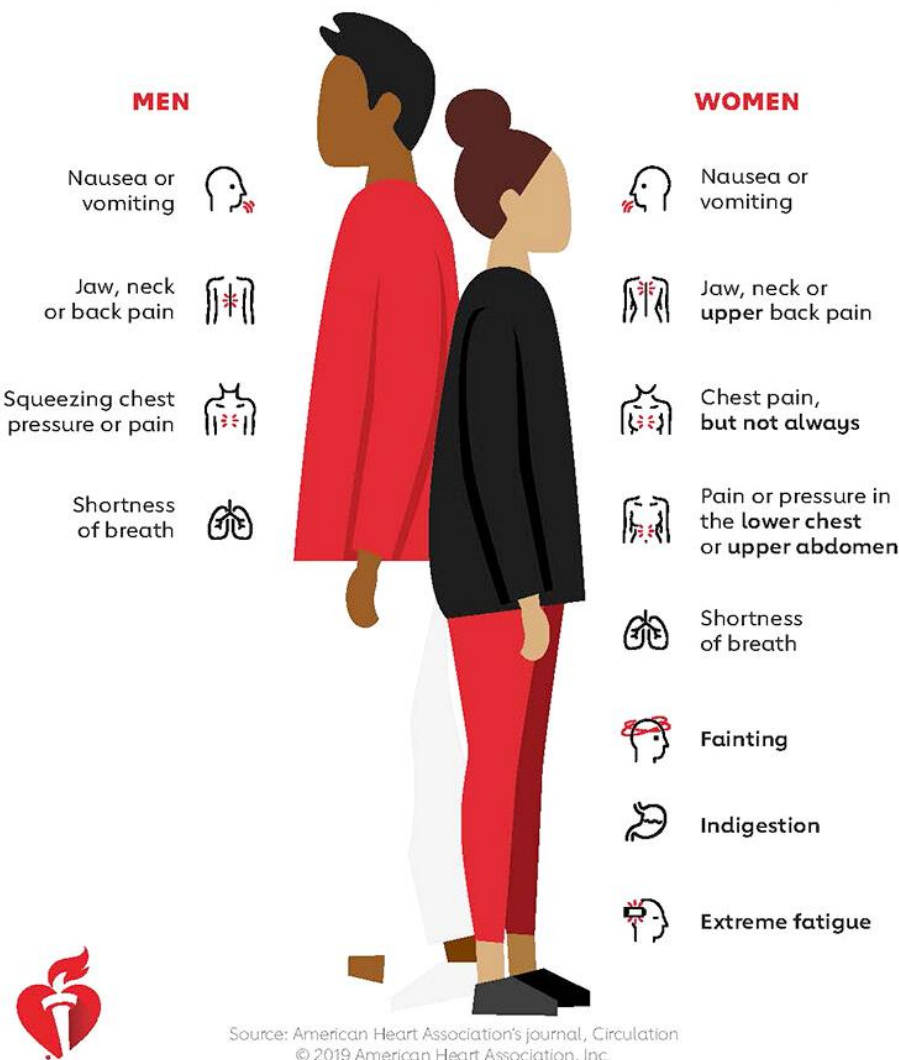
Bias should be mitigated based on the task



Gender in loan application can be discriminatory

Gender in health application can be necessary

HEART ATTACK SYMPTOMS: MEN VS. WOMEN

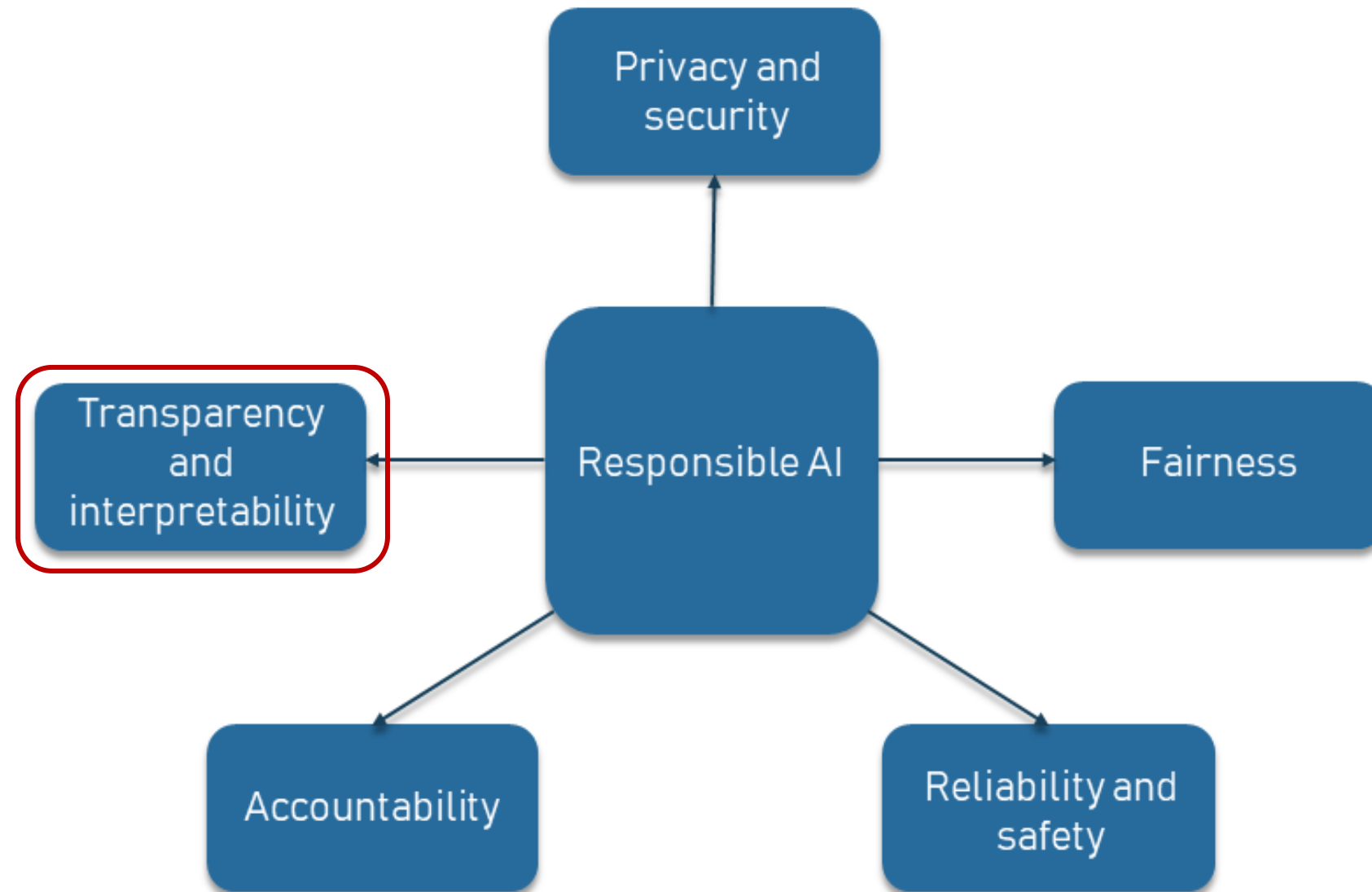


Responsible AI

Despite the exponential increase in the amount of machine learning models in production, only a few companies have dedicated the time and effort to ensure these models are deployed responsibly. This is partly because there is no standard definition of Responsible AI.

Responsible AI is the practice of designing, developing, and deploying AI with good intention to empower employees and businesses, and fairly impact customers and society—allowing companies to engender trust and scale AI with confidence. (AI Ethics Governance, Accenture)

KEY PRINCIPLES OF RESPONSIBLE AI



Black box vs white box models

White-box models are the type of models which one can clearly explain how and why they produce certain predictions.

Essentially, the relationship between the data used to build the model and the model outcome is explainable.

E.g. Linear Regression, Logistic Regression

Conversely, **Black-box** models are more advanced in approximating more complex relationships between the input data & the target, but lack-clarity.

E.g. Deep Neural Networks, Random Forest



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White box model



=

Black box model

Both cars can drive you from Point A to Point B

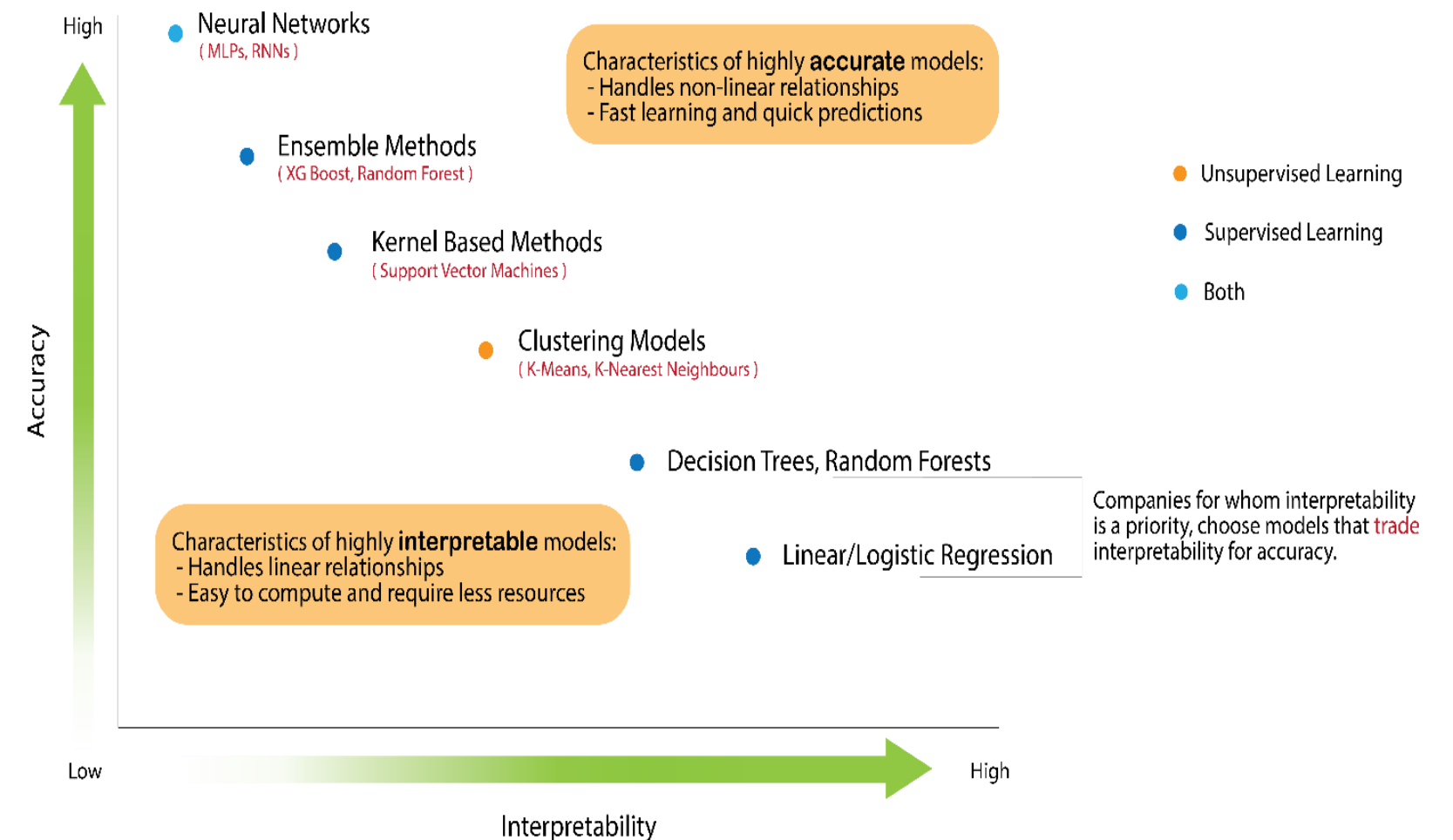
★ ★ ★	Good for small errands Handles small and less complex data	★
★	Reliability with complex terrains Handles data complexity	★ ★ ★
★	Carries more passengers Can handle large amounts of data	★ ★ ★
	Uses more resources Requires more computing power	★ ★ ★

So, what's the issue? Why don't we use Black-box models?

Algorithms like Neural Networks, Ensemble models can be highly accurate but not very interpretable. While, models like Linear Regression and Logistic Regression that are highly interpretable but may not be as accurate.

In some situations, we may be comfortable with this **trade-off** between model accuracy and interpretability.

For example, we might not care why a model recommends watching a particular movie to someone, but we do care why someone has been rejected for a credit card application. And in the later case you need to be both **accurate** and **transparent**.



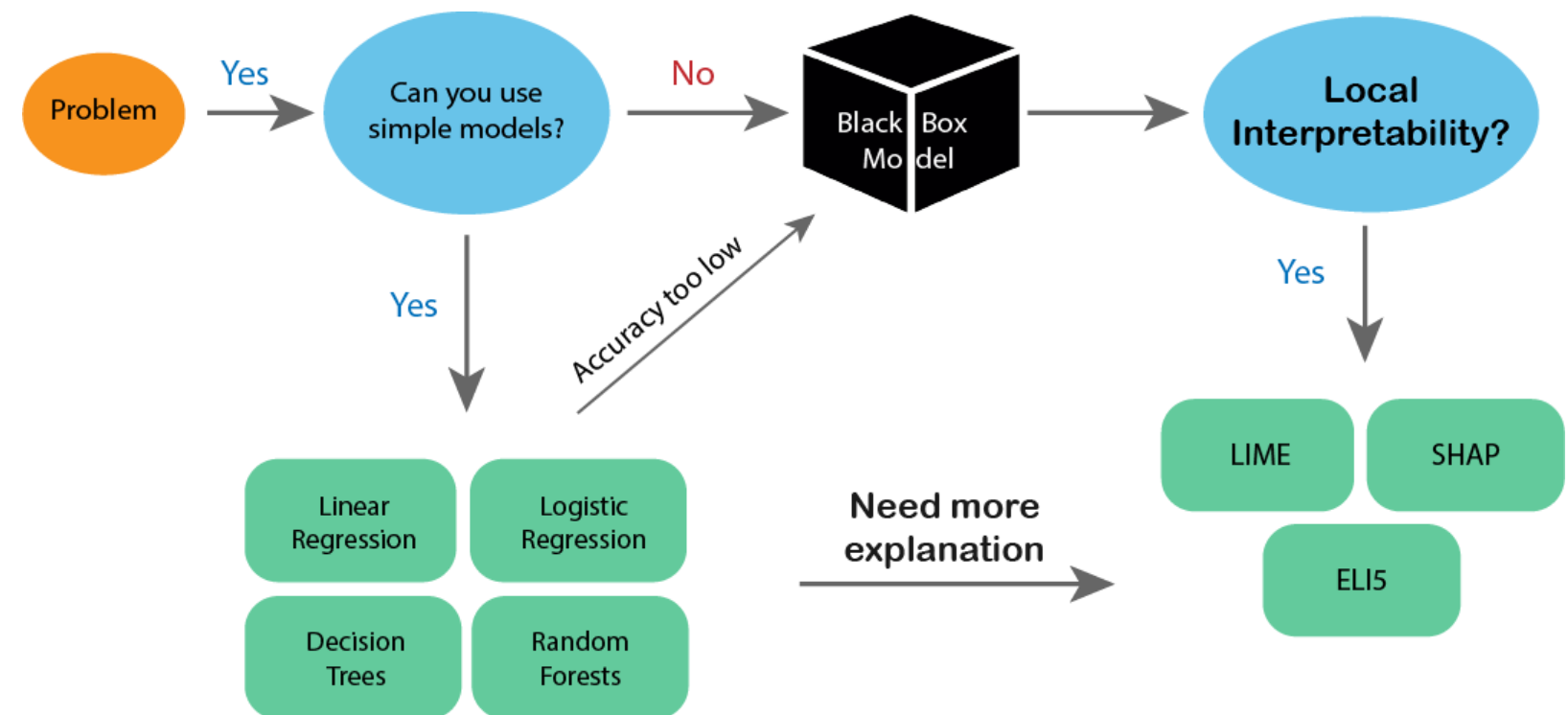
Some things you just can't describe, no matter how hard you try!

Models like **BERT**, **GPT-3** are deep learning models that give state-of-the-art results on a wide variety of Natural Language Processing tasks. They are trained on **340M** and **176B** features respectively.

Explainable AI Solutions and how they can help?

Without this technique, the second best-option is to compromise on the accuracy and rely on a model that is generally explainable.

This **novel approach** allows to keep the complexity without compromising the interpretability of the model.

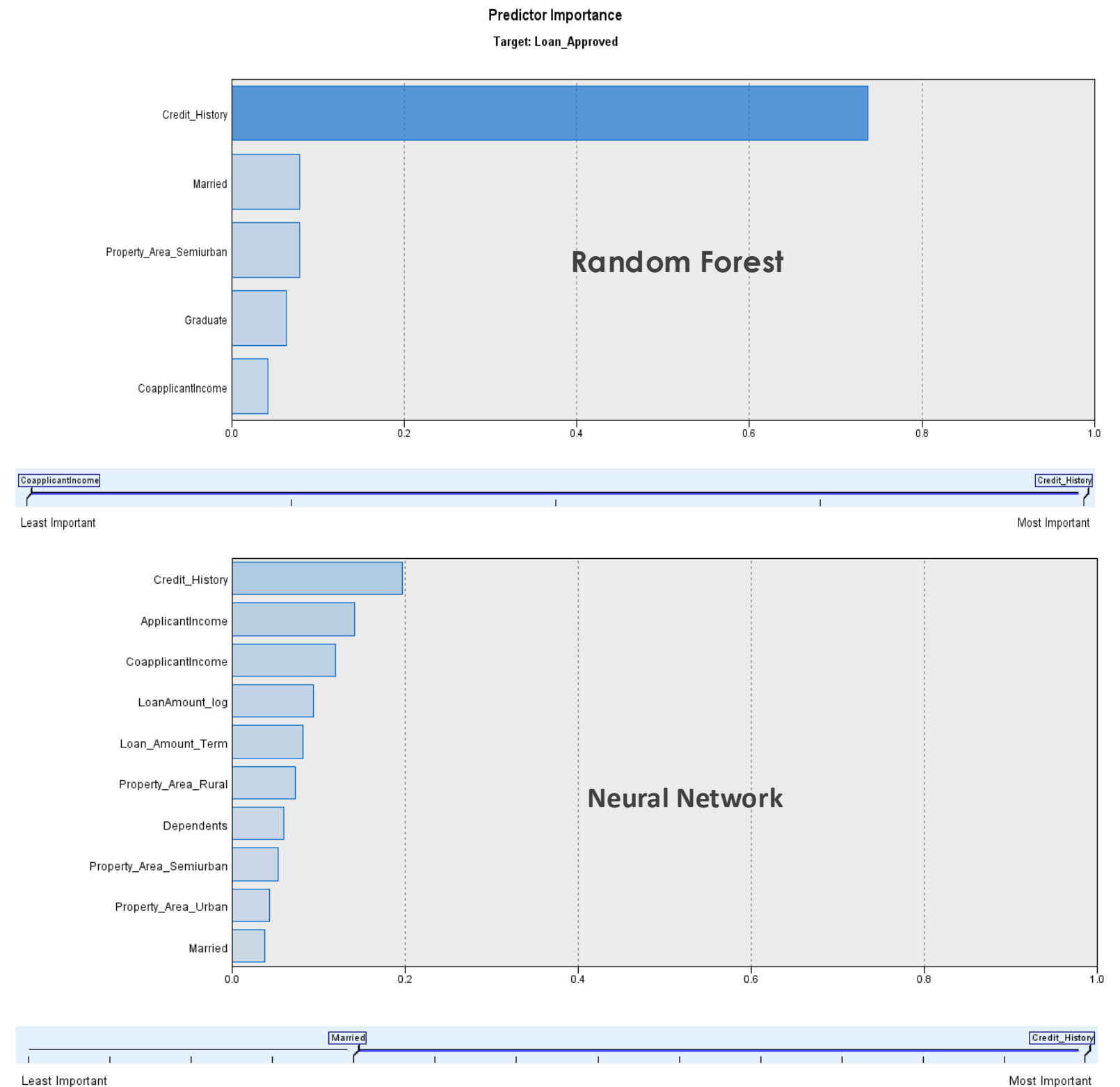


Available Solutions - Global Interpretability

Currently we have global interpretability (which explains the entire model behavior but not an individual prediction)

Also, these indicators differ with implementation of different models and they are true for the model as a whole, but not every individual prediction is based on these rankings.

More importantly, these charts do not tell us whether the variables effected the outcome positively or negatively.



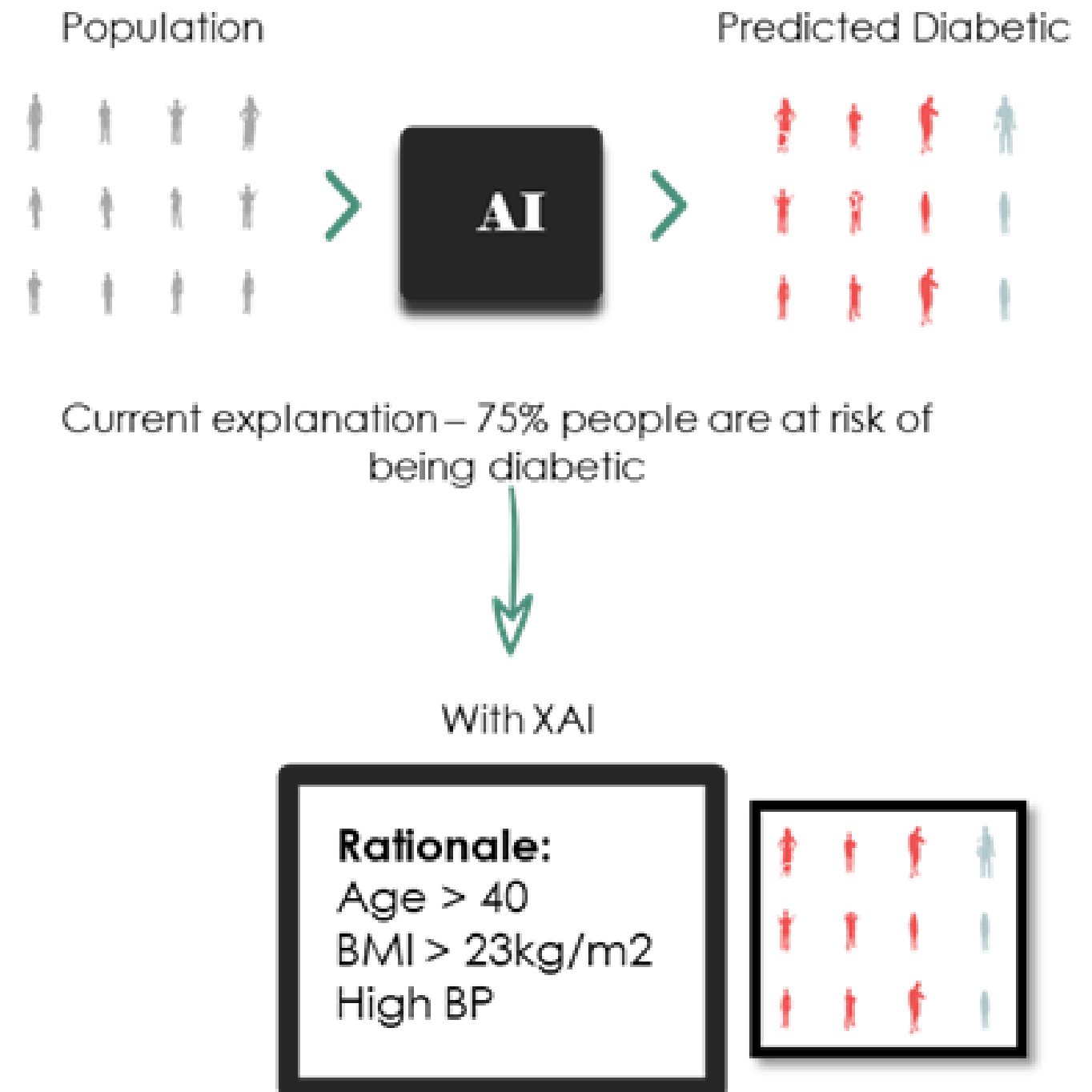
Explainable Ai - XAI

Let' understand this with an example:

Consider an algorithm which is classifying a population as diabetic and healthy.

Without XAI solutions, we do not understand the rationale behind the model making the classification.

With XAI, we not only understand the rational behind the outputs but we are also able to uncover it at an individual level.



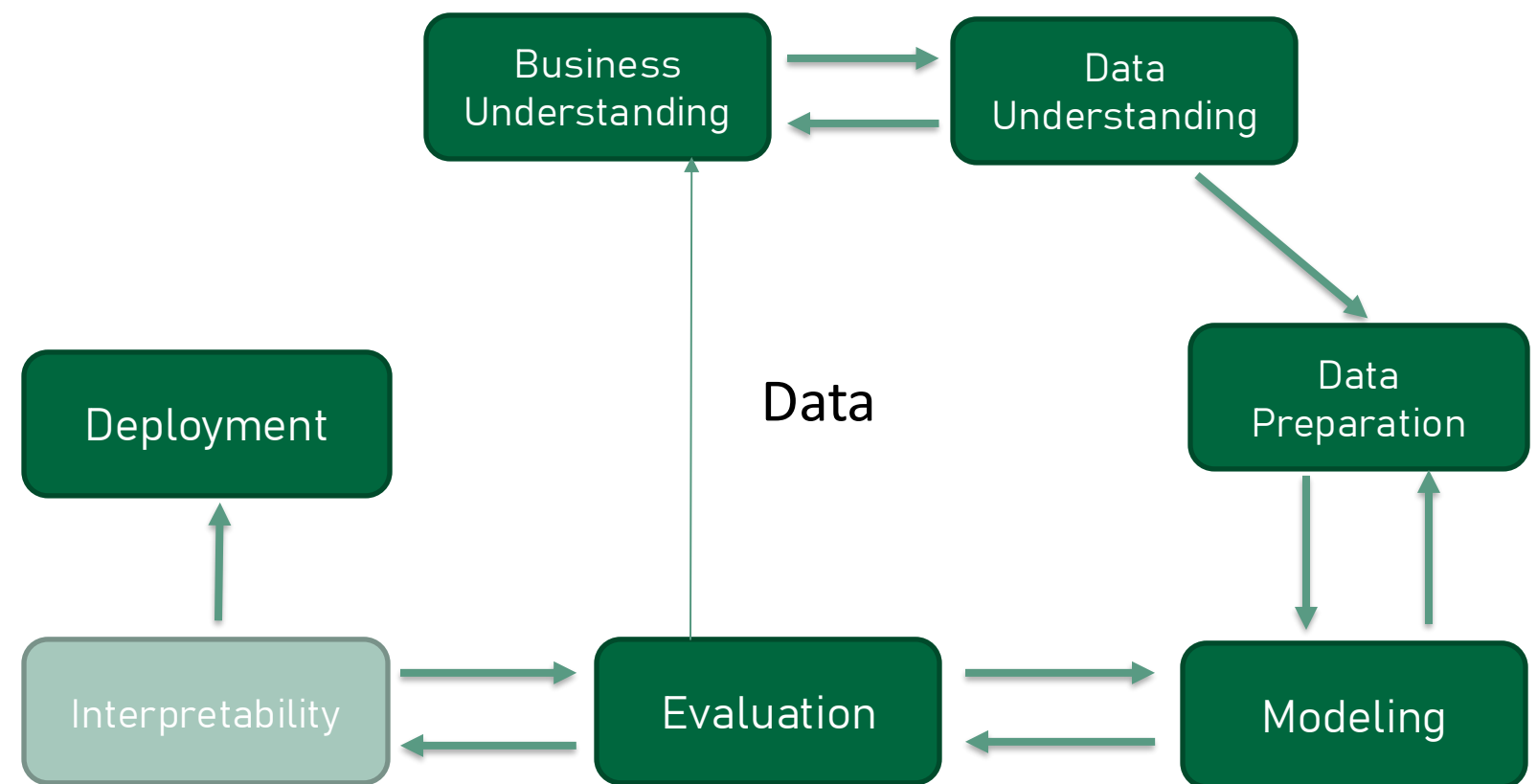
Integrating with current processes

Model-agnostic methods can be used on any machine learning model and are applied after the model has been trained and evaluated.

They answer two very important questions,

1. How does the model work
2. Why did the model make certain prediction

CRISP – DM



Thank you!